Hyperspectral Image Classification With Mamba

Zhaojie Pan, *Student Member, IEEE*, Chenyu Li, *Student Member, IEEE*, Antonio Plaza, *Fellow, IEEE*, Jocelyn Chanussot, *Fellow, IEEE*, and Danfeng Hong, *Senior Member, IEEE*

***Abstract*— Local and global spectral and spatial information is crucial for hyperspectral image (HSI) classification. However, modeling the global context has been challenging due to the limitations of receptive fields and quadratic complexity. Mamba's ability to leverage long-range dependencies with linear computational complexity offers an effective approach to alleviate this issue; however, it does lead to the loss of local detail information. To address this challenge, we propose a novel local-to-global Mamba for HSI classification, termed MambaLG. MambaLG consists of a dual-branch strategy, comprising two core modules: a local and global spatial modeling module (SpaM) and a short- and long-range spectral dynamic perception module (SpeM). In the SpaM, the local and global spatial information is sequentially extracted and integrated, aiming to capture global spatial semantics while preserving the integrity of local 2-D spatial structures. In the SpeM, we utilize local spectral extraction, spectral grouping, and spectral dynamic correlation clustering (SDCC) modules, leveraging Mamba's strengths in exploring long-range dependencies for more precise short- and long-range spectral feature modeling. Additionally, we introduce a gate attention unit into MambaLG and design a more efficient and interpretable manner for merging spatial and spectral features. Experimental results across multiple datasets (encompassing urban and agricultural scenes) indicate that MambaLG surpasses state-of-the-art algorithms regarding classification accuracy (CA) and inference speed. Comprehensive ablation studies substantiate the advantages of MambaLG in modeling local and global spatial context, enhancing short- and long-range spectral perception, and fusing spatial and spectral information. The codes will be openly available at https://github.com/danfenghong/IEEE_TGRS_MambaLG to facilitate the reproduction of experimental results.**

***Index Terms*— Decoupling and fusion, hyperspectral image (HSI) classification, local and global spatial context, Mamba, short- and long-range spectral dependencies.**

I. Introduction

HYPERSPECTRAL images (HSIs) are renowned for their ability to capture intricate spectral features of terrestrial objects, thereby offering a remarkable advantage in the precise identification of various ground features. This capability has positioned HSIs as an indispensable tool in a multitude of sectors such as agricultural preservation [1], urban development [2], mineral mapping [3], and environmental protection [4]. The classification of HSIs is pivotal for their application in various fields.

An HSI represents a 3-D data structure that captures the spatial distribution characteristics in the spatial dimension and encodes the spectral properties of materials in the spectral dimension. Early HSI classification algorithms predominantly focused on the utilization of spectral information [5], which resulted in their inherent challenge to address the hyperspectral–heterospectral issue [6]. Fortunately, the integration of spectral and spatial information provided an effective solution to the problem [7]. However, traditional classification methods (which rely heavily on handcrafted features) often struggle when applied to complex scenes due to their limited robustness and generalizability. In contrast, deep learning (DL)-based algorithms can extract deep, nonlinear features and effectively tackle these challenges [8]. Nevertheless, as the utilization of HSIs becomes increasingly prevalent in both civilian and military applications, researchers have identified the need for more sophisticated geospatial data-mining techniques to address the complexities inherent to such environments adequately. Consequently, various DL-based approaches have been proposed for more precise utilization of spatial information [9], [10].

Local and global are two crucial types of spatial information, and DL-based HSI classification methods can be categorized according to the different ways in which they exploit these two types of information, namely local learning and global learning methods. Local learning methods, also known as patch-based methods, adeptly capture local spatial details by examining the spatial correlation within a patch [11]. In [12], convolutional neural network (CNN)-based methods are designed to model the local spectral and spatial information. In [13]. Here, the attention mechanism is introduced into HSI classification to exploit key features. To extract long-range dependencies, transformer-based methods have been adopted [14]. Although such patch-based methods achieve good performance, they are constrained by the framework's inability to consider global features, which restricts their performance.

In contrast, global learning methods—known as fully connected (FCN)-based methods—effectively resolve the problem of patch-based methods (particularly, regarding their insufficient receptive field) by taking the whole HSI as

Received 19 October 2024; revised 1 December 2024; accepted 19 December 2024. Date of publication 23 December 2024; date of current version 3 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42271350 and Grant 42241109, and in part by the International Partnership Program of the Chinese Academy of Sciences under Grant 313GJHZ2023066FN. *(Corresponding author: Chenyu Li.)*

Zhaojie Pan and Chenyu Li are with the School of Mathematics, Southeast University, Nanjing 211189, China (e-mail: chenyuli.erh@gmail.com).

Antonio Plaza is with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica, University of Extremadura, 10003 Cáceres, Spain (e-mail: aplaza@unex.es).

Jocelyn Chanussot is with INRIA, 38330 Montbonnot-Saint-Martin, France (e-mail: jocelyn.chanussot@inria.fr).

Danfeng Hong is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: hongdf@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2024.3521411

1558-0644 © 2024 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

input [15], [16]. In [17], spectral and spatial feature extraction techniques with distinctive characteristics are designed, effectively enhancing the model's performance. In [18], global spectral and spatial learning is introduced into wetland HSI classification. However, the aforementioned methods mainly rely on the CNN architecture for feature extraction, which makes it difficult for them to effectively leverage the advantages of the global framework. To address this problem, transformer-based global HSI classification methods have been developed. Although such methods are capable of modeling global dependencies, their computational complexity makes their application to images with large spatial scales difficult.

Recently, the selective state space model (SSM) (Mamba) [19] has been introduced to the field of remote sensing [20], [21], offering the capability to model long-range information with linear complexity. The feature of Mamba significantly reduces the computational strain associated with the extraction of global spatial information, thereby facilitating the precise modeling of complex spatial information. Furthermore, Mamba's state transfer mechanism adeptly exploits the long-range dependency, which provides a novel perspective for the model to discern the spectral difference and their intrinsic correlation. Thus, Mamba exhibits considerable potential in HSI processing, particularly when compared to methods reliant on the transformer and CNN.

Although Mamba possesses the aforementioned advantages, the following issues need to be taken into account when employing it for the HSI classification.

1) The global learning framework overlooks the exploitation of local spatial details. Mamba's 1-D linear scanning mechanism efficiently models global spatial semantics with linear complexity; however, when using this mechanism to extract 2-D spatial information, the 2-D data needs to be converted into a 1-D sequence. Such a transformation may spatially disconnect adjacent pixels in the sequence, potentially compromising the integrity of the local spatial structure.
2) The fusion of spatial and spectral information is as critical as the extraction process itself. Given the integrated nature of spatial and spectral features in HSIs, accurate identification of ground objects requires a joint representation of spectral and spatial features. However, existing methods struggle to achieve a good balance between feature extraction and feature fusion.

To address the aforementioned challenges, this article proposes a new local-to-global framework (called MambaLG) for HSI classification. Our key introspection is that the expected MambaLG for an ideal spatial and spectral feature extraction system should be both local detail-friendly and global context-friendly, as summarized in Fig. 1. MambaLG facilitates the synergistic extraction and sophisticated integration of spatial and spectral information through two elaborate feature extraction modules and a novel fusion approach. The local and global spatial modeling module (SpaM) and the short- and long-range spectral dynamic perception module (SpeM) are pivotal components of MambaLG. SpaM incorporates CNN- and Mamba-based extraction strategies that capture both local and global spatial information, ensuring efficient linear computational complexity (while preserving 2-D local spatial details). SpeM is designed to evaluate the importance of different bands for the final classification through local spectral extraction, spectral grouping, and spectral dynamic correlation clustering (SDCC) to model short- and long-range spectral correlations. Finally, inspired by the gated architecture of mamba, we introduce a gated attention paradigm into MambaLG, thereby achieving the fusion of spectral and spatial information through a functional-level interaction. The primary contributions of this article can be summarized as follows.

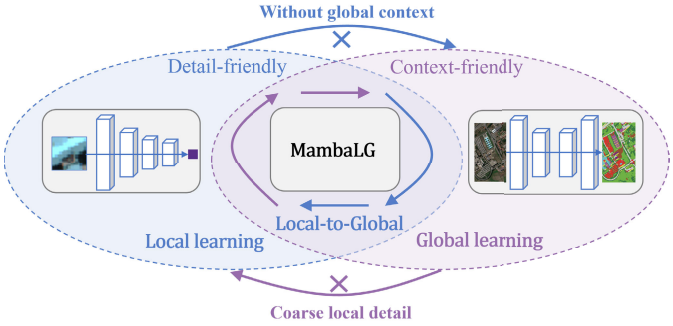


Fig. 1. Visual comparison of HSI classification frameworks. The local learning framework is detail-friendly, but it cannot exploit the global context. The global learning framework is global context-friendly, but it cannot model the local details. The local-to-global mamba inherits the strengths of local and global learning frameworks while mitigating their weaknesses.

1) A novel local-to-global framework, MambaLG, is proposed for HSI classification. MambaLG is capable of fully extracting and integrating local and global spatial and spectral information through SpaM, SpeM, and gated attention units.
2) To make full use of both local and global 2-D spatial information, the SpaM is proposed to leverage Mamba's strengths in modeling global dependencies while preserving the integrity of local 2-D spatial structures.
3) To balance the extraction and fusion of spatial and spectral information, the gated attention paradigm is introduced for achieving channel-level fusion of global spectral and spatial information by a form of functional interaction.

The remainder of this article is organized as follows. In Section II, we provide an overview of related work. In Section III, the proposed method is described in detail. Section IV describes our experimental results and parameter analyses. Finally, in Section VI, we summarize our contributions and conclude the article with some remarks.

II. Related Work

A. Mamba and Its Application to HSI Classification

Mamba [19], originating from the SSM in cybernetics, has shown the potential to replace transformers in various visual tasks. The traditional SSM is a continuous system [22], and its fundamental formula is

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h'(t) + \mathbf{D}x(t) \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

where $\mathbf{A} \in \mathbb{R}^{d\times d}$, $\mathbf{B}$, $h'(t)$, $h(t) \in \mathbb{R}^{d\times 1}$, $\mathbf{C} \in \mathbb{R}^{1\times d}$, $\mathbf{D} \in \mathbb{R}$. To be effectively utilized in DL applications, SSM is transformed into a discrete version through zero-order hold discretization [23]. The discrete SSM (S4) can be represented as

$$h_i = \overline{\mathbf{A}} h_{i-1} + \overline{\mathbf{B}} x_i \tag{3}$$

$$y_i = \mathbf{C} h_i + \mathbf{D} x_i \tag{4}$$

where i is denotes the position of the current state within the sequence, $\overline{\mathbf{A}} = \exp(\triangle A)$, $\overline{\mathbf{B}} = (\triangle \mathbf{A})^{-1}(\exp(\triangle \mathbf{A} - I)) \cdot \triangle B \approx \triangle B$. From (3) and (4), we can infer that sequence information can achieve global interaction with linear complexity through the propagation of h_i.

Building upon its foundation, Mamba elevates the functionality of B, C, and $\triangle$ by redefining them as functions of x. This transformation empowers Mamba to efficiently select data in an input-dependent manner. The selective SSM (S6) endows Mamba with the capability to dynamically adjust its mappings in response to fluctuating inputs, ensuring that the model achieves promising results across multiple visual tasks [24], [25], [26]. In [27], Mamba demonstrated its advantages in remote sensing semantic segmentation.

Recently, several Mamba-based frameworks for HSI classification have been introduced, achieving remarkable performance [28]. In [29], a novel spectral–spatial Mamba network is proposed, which employs a dual-branch framework to simultaneously leverage both spectral and spatial information. In [30], a centralized Mamba-Cross-Scan mechanism is designed to transform patch-level images into sequential data, thereby enhancing the model's ability to model local spatial information. While these approaches have significantly enhanced the capabilities of Mamba-based frameworks in HSI classification, they are still limited to patch-based local learning, which presents challenges in the task of leveraging global spatial information. In [31], MambaHSI is proposed to model global spectral and spatial features, but it does not fully address the limitations of the global learning framework in extracting local spatial structures.

B. Spectral–Spatial HSI Classification Methods

DL-driven spectral–spatial HSI classification methods can be divided into two categories: 3-D methods [32] and spectral–spatial decoupling methods [10], [16], [33]. Three-dimensional methods maintain the integrated spectral–spatial nature of HSIs, facilitating the extraction of spectrally and spatially correlated information. Nevertheless, 3-D methods can be less reliable in complex environments because of the intricate nonlinear relationships between spectral and spatial features. Fortunately, decoupling-based methods use separate modules to independently extract spectral and spatial features, enabling more efficient exploitation of the most discriminative characteristics.

Decoupling-based methods can be categorized into dual-branch-based and cascade-based approaches. Cascade-based methods extract information by connecting spatial and spectral feature extraction modules serially [34]. However, the processing carried out by one module can impact the processing of the subsequent module, which leads to difficulties in fully exploiting spectral or spatial information. In contrast, dual-branch-based methods process spectral and spatial features in parallel, effectively avoiding the aforementioned problem. In [35], the techniques for extracting spectral and spatial information are integrated into the module, leading to outstanding performance.

The success of decoupling-based HSI classification methods depends on a balanced strategy that includes both the decoupling of distinct features and their comprehensive fusion. To integrate spectral and spatial information in the decoupled framework, researchers have proposed numerous fusion methods, which can be categorized into midprocessing fusion and postprocessing fusion. Postprocessing fusion is the most popular approach, but it results in limited interaction between the spatial and spectral features, which is not conducive to the fusion of the two features. Midprocessing fusion enhances the coupling of spectral and spatial information by strengthening the information flow between their respective extraction techniques [18]. Nevertheless, these approaches may inadvertently lead to information interference, impairing the network's effectiveness in decoupling spatial and spectral information. To address these challenges, this article introduces novel strategies for decoupling and fusion (specifically, functional-level interactive fusion), ensuring both the separation and intelligent integration of spectral and spatial features.

III. Methodology

We propose a novel Mamba-based method for local-to-global HSI classification (called MambaLG). Our MambaLG utilizes the SpaM to model local and global spatial information and employs the SpeM to generate spectral weights by perceiving short- and long-range interband correlation information. It integrates the dual-branch architecture with the gated attention mechanism, thereby achieving the synergistic fusion of spatial and spectral information. In this section, we introduce the overall framework of our MambaLG, provide detailed descriptions of the SpaM and SpeM, and conclude with an explanation of the objective function design.

A. Overall Architecture

Our MambaLG employs a dual-branch architecture, with the SpaM and SpeM at its core, as shown in Fig. 2. The SpaM encompasses a multiscale spatial information extraction module (MSpaFE), a local spatial correlation extraction (LSCE) module, and a global spatial context modeling module (GSCE). The SpeM includes a multiscale spectral information extraction module (MSpeFE), a spectral grouping module, and an SDCC module.

Let $F \in \mathbb{R}^{H\times W\times C}$ denote the input HSI, where H, W, and C represent the number of rows, columns, and spectral bands, respectively. First, we utilize a feature extraction module composed of 1×1 2-D convolution, batch normalization (BN), and Gaussian error linear unit (GELU) to extract shallow features. This process can be represented as

$$F_s = \text{In_head}(F) \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

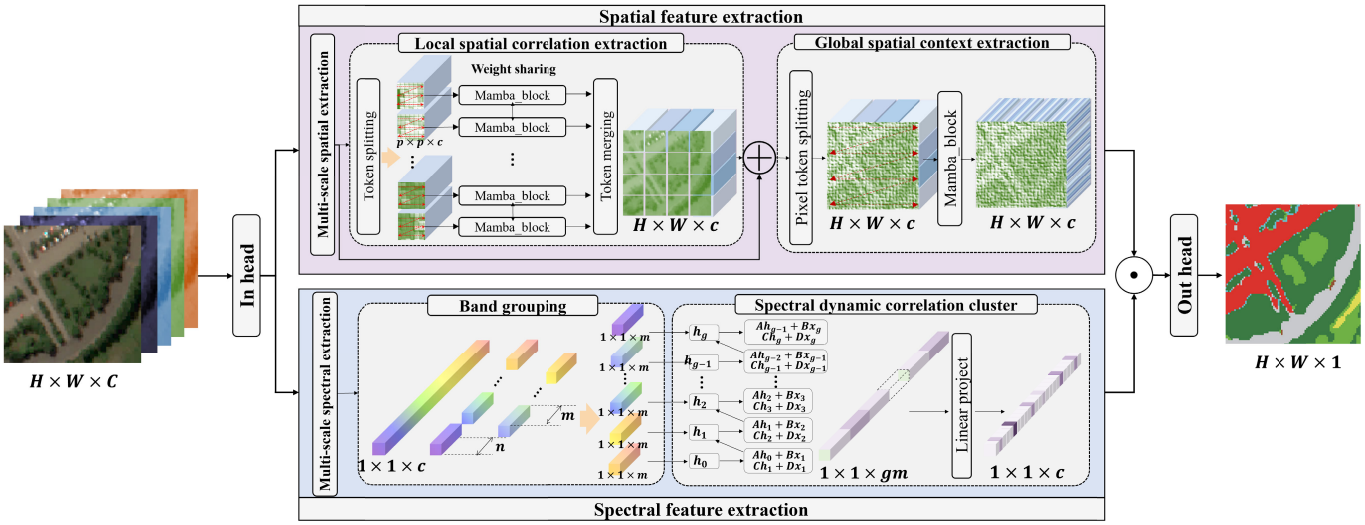


Fig. 2. Flowchart of the proposed MambaLG for HSI classification.

where In_head(.) is the shallow feature extraction module, $F_s \in \mathbb{R}^{H\times W\times c}$ is the shallow feature, and c is the number of channels of F_s. Then, we utilize the SpaM and SpeM separately to extract spectral and spatial information in a decoupled manner.

In the SpaM, we first utilize the MSpaFE (MSFE) to consolidate information across different local spatial scales. Following this, the LSCE is employed to identify local spatial correlation from the features. Subsequently, we integrate the multiscale spatial information with the local spatial correlation using skip connections. This comprehensive local spatial structure modeling process is mathematically formulated as follows:

$$F_{ms}^{\text{spa}} = \text{MSFE}_{\text{spa}}(F_s) \tag{6}$$

$$F_{\text{spa}}^{l} = \text{LSCE}\left(F_{ms}^{\text{spa}}\right) + F_{ms}^{\text{spa}} \tag{7}$$

where $F_{ms}^{\text{spa}} \in \mathbb{R}^{H\times W\times c}$ and $F_{\text{spa}}^{l} \in \mathbb{R}^{H\times W\times c}$ are the outputs of MSFE and LSCE, respectively. $\text{MSFE}_{\text{spa}}(.)$ and LSCE(.) denote the feature processing carried out by MSFE and LSCE, respectively. Then, the GSCE is designed to extract global spatial semantics. Hence, the output of GSCE $F_{\text{spa}} \in \mathbb{R}^{H\times W\times c}$ can be obtained as follows:

$$F_{\text{spa}} = \text{GSCE}\left(F_{\text{spa}}^{l}\right) \tag{8}$$

where GSCE(.) is the feature extraction function of GSCE.

In the SpeM, we first employ the MSpeFE to facilitate the initial modeling and integration of multiscale spectral information. Following this, we utilize a spectral grouping module to perform coarse clustering of spectral information. This process is succinctly captured by

$$F_{ms}^{\text{spe}} = \text{MSFE}_{\text{spe}}(F_s) \tag{9}$$

$$X = \text{Group}_{m,n}\left(F_{ms}^{\text{spe}}\right) \tag{10}$$

where $\text{MSFE}_{\text{spe}}(.)$ denotes the function of MSpeFE and $F_{ms}^{\text{spe}} \in \mathbb{R}^{1\times c}$ is the output of MSpeFE. $\text{Group}_{m,n}(.)$ is the spectral grouping module, where m is the number of bands in each band group and n denotes the number of bands intervening between the starting bands of two adjacent band groups. $X = [x_1, x_2, x_3, \ldots, x_i, \ldots, x_g]^T \in \mathbb{R}^{g\times m}$ is the band group sequence, where g is the number of band groups. Subsequently, we utilize Mamba to perform a detailed analysis of spectral information within the band group sequence, aiming to model the dynamic variations in global spectral information along the channel axis. The processing of X by the SDCC is articulated by the following equation:

$$F_{\text{spe}} = \text{SDCC}(X) \tag{11}$$

where $F_{\text{spe}} = [w_1, w_2, \ldots, w_i, \ldots, w_c] \in \mathbb{R}^{1\times 1\times c}$ is the processing outcome of SDCC.

Then, we achieve the coupling of channel-level spectral and spatial information through an elegant channel-wise multiplication operation

$$F_{\text{fusion}} = F_{\text{spa}} \odot F_{\text{spe}} \tag{12}$$

where $F_{\text{fusion}} \in \mathbb{R}^{H\times W\times c}$ and $\odot$ represents element-wise multiplication. According to (12), the outputs of the SpeM act as weights for F_{spa}, thus allowing the spectral branch to enhance the spatial branch. By employing a novel functional interaction approach, our MambaLG maintains the correlation between spectral and spatial information even when they are extracted independently in a decoupled fashion, leading to a more effective and interpretable integration of spatial and spectral information.

Finally, we utilize an output module that consists of a 3×3 average pooling layer, followed by channel-wise convolution, and then apply the argmax function to determine the category of each pixel in the image. It can be represented as

$$F_{\text{out}} = \text{Con}_{1\times 1}\left(\text{Avgpool}_{3\times 3}(F_{\text{fusion}})\right) \tag{13}$$

$$Y = \text{argmax}(F_{\text{out}}) \tag{14}$$

where $Y \in \mathbb{R}^{H\times W}$ and $F_{\text{out}} = [p_1, p_2, \ldots, p_k, \ldots, p_K] \in \mathbb{R}^{H\times W\times K}$ is the output of MambaLG, with $p_k \in \mathbb{R}^{H\times W}$ representing the probability that the pixel at the corresponding position in the image belongs to the kth class (K denotes the total number of classes).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

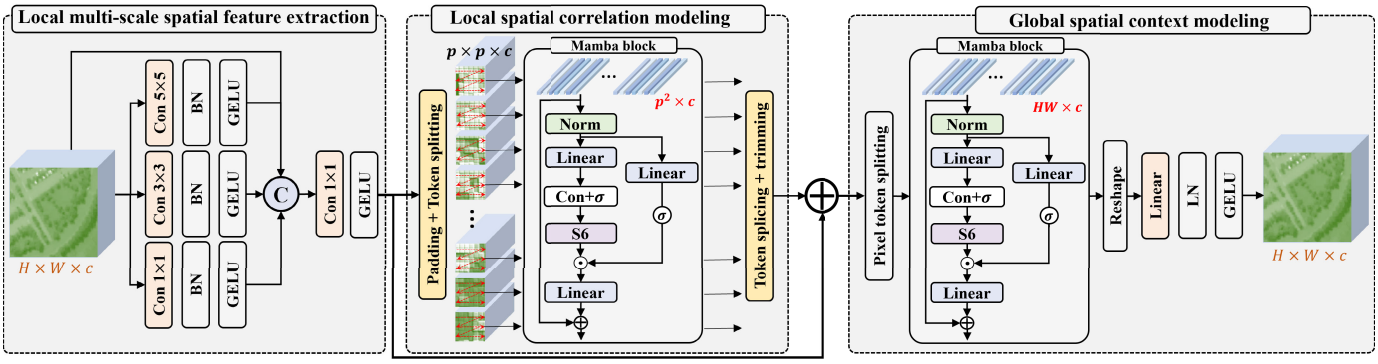


Fig. 3. Graphical illustration of the SpaM for local-to-global spatial semantic modeling.

B. Local-to-Global Spatial Semantic Modeling

Integrating and utilizing local and global spatial information facilitates the enhancement of the model's capacity to recognize land-cover features with complex spatial structures. Moreover, the precision of extracting local spatial information is conducive to the modeling of global spatial information. Subsequently, the SpaM adopts a cascaded approach, starting with local spatial structure explosion and then global spatial context excavation for global spatial information modeling. This section will specifically introduce the process through which SpaM, based on Mamba, accomplishes the joint utilization of local and global spatial information. The detailed structure of SpaM is shown in Fig. 3.

1) Local Multiscale Spatial Feature Extraction: Mamba is designed to handle 1-D sequential information, which may undermine the 2-D spatial structure [24]. To address this issue, we design the MSFE to model multiscale spatial information for the protection of local spatial structure.

The MSFE employs a 3 × 3.2-D convolution to extract information from adjacent pixels, a 5 × 5 average pooling to capture medium-scale rotation-invariant features, and a 1 × 1.2-D convolution to refine pixel-level details. The multiscale feature extraction process can be described as follows:

$$F_1^{\text{spa}} = \text{GELU}(\text{BN}(\text{Con}_{3\times3}(F_s))) \tag{15}$$

$$F_2^{\text{spa}} = \text{GELU}(\text{BN}(\text{Avgpool}_{5\times5}(F_s))) \tag{16}$$

$$F_3^{\text{spa}} = \text{GELU}(\text{BN}(\text{Con}_{1\times1}(F_s))) \tag{17}$$

where F_1^{spa}, F_2^{spa}, and F_3^{spa} have the same spatial size as that of F_s. Subsequently, the multiscale features are concatenated with the original features and the spatial information of each pixel is dynamically integrated through 1 × 1.2-D convolution. We can formulate the process as

$$F_{ms}^{\text{spa}} = \text{GELU}\big(\text{Con}_{1\times1}\big(\text{Cat}\big(F_1^{\text{spa}}, F_2^{\text{spa}}, F_3^{\text{spa}}, F_s\big)\big)\big). \tag{18}$$

2) Local Spatial Correlation Modeling: To facilitate further aggregation of local spatial information, the LSCE module is designed. To accommodate this nonoverlapping segmentation for images with diverse spatial sizes, we first apply an expansion technique. Following this, we segment the HSI into patches of uniform size and employ the Mamba block for the correlated modeling of local spatial patches.

Assume that $p \times p$, $H \times W$, and $h \times w$ are the sizes of the patch, original image, and the expanded image, respectively, then $h = H + H\text{mod}p$, $w = W + W\text{mod}p$, and the number of patches is $((h \times w)/p^2)$. Hence, the segmented HSI after segmentation can be expressed as $P = [P_1, P_2, \ldots, P_i, \ldots, P_{(h\times w)/p^2}]$, where $P_i \in \mathbb{R}^{p\times p\times c}$. We can formulate the processing of the Mamba block as

$$P_{i,x}, P_{i,z} = \text{chunk}(\text{linear}(\text{LN}(\text{flatten}(P_i)))) \tag{19}$$

$$P_{i,x}^m = \text{S6}(\text{Silu}(\text{Conv}(P_{i,x}))) \tag{20}$$

$$P_{i,\text{out}} = \text{Reshape}\big(\text{linear}\big(\text{LN}\big(P_{i,x}^m\big) \odot \text{Silu}\big(P_{i,z}\big)\big) + \text{flatten}(P_i)\big) \tag{21}$$

where Conv(.) is a 1-D convolution, LN(.) stands for layer normalization (LN), Silu(.) is the sigmoid linear unit (Silu), $P_{i,x}$, $P_{i,z}$, and $P_{i,x}^m \in \mathbb{R}^{p^2\times2c}$ are the intermediate variables, and $P_{\text{out}} = [P_{1,\text{out}}, P_{2,\text{out}}, \ldots, P_{i,\text{out}}, \ldots, P_{(h\times w)/p^2,\text{out}}]$ is the result generated by the LSCE, with $P_{i,\text{out}} \in \mathbb{R}^{p\times p\times c}$. Note that the Mamba block for each patch conducts weight-sharing to effectively restrain noise and acquire local correlation characteristics in accordance with global laws.

Subsequently, we successively split the patches and reconstructed the image. Given that the padding operation modifies the original dimension of the image, we ultimately trim off the padded portion. The final output of LSCE $F_{\text{spa}}^{l,m}$ can be obtained by

$$F_{\text{spa}}^{l,m} = \text{merging}(P_{\text{out}}) \tag{22}$$

where merging(.) denotes the processing related to token splicing and trimming.

3) Global Spatial Context Modeling: Mamba can model long-range dependencies with linear complexity, effectively alleviating the limitations of CNNs and transformers related to their receptive fields and computational complexity. Hence, we use Mamba to exploit the global spatial features of the HSIs. Specifically, F_{spa}^l is first segmented (in pixel-by-pixel fashion) to generate a 1-D sequence $T = [T_1, T_2, \ldots, T_i, \ldots, T_{H\times W}]^T \in \mathbb{R}^{HW\times c}$. Then, the sequence is fed into the Mamba block for modeling global spatial information. It can be formulated as

$$T_x, T_z = \text{chunk}(\text{linear}(\text{LN}(T))) \tag{23}$$

$$T_x^m = \text{S6}(\text{Silu}(\text{Conv}(T_x))) \tag{24}$$

$$T_{\text{out}} = \text{Reshape}(\text{linear}(\text{LN}(T_x^m) \odot \text{Silu}(T_z)) + T) \tag{25}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

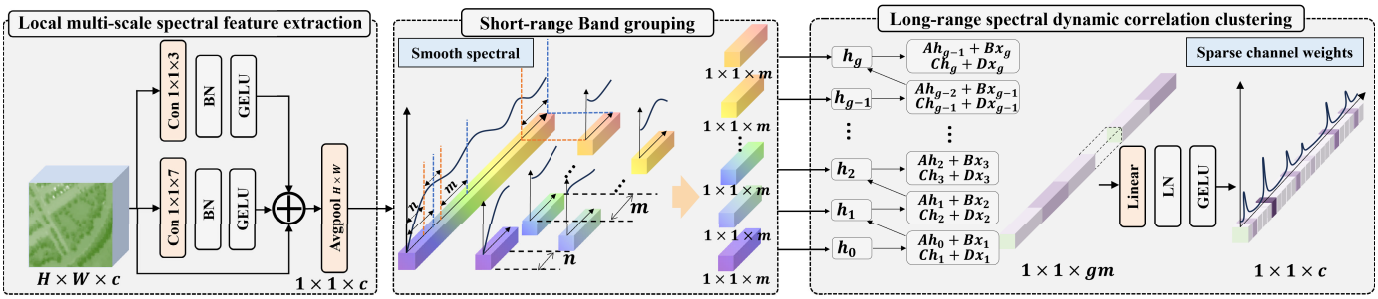


Fig. 4. Illustration of the SpeM for short- and long-range spectral dynamic perception.

where the $T_{i,x}, T_{i,z}, T_x^m$ are the intermediate variables of the global Mamba block and $T_{\text{out}} \in \mathbb{R}^{H\times W\times c}$ is the output of the block.

Finally, we use an information integration module composed of a linear layer, a layer normalization, and an activation layer to further process the spatial information in preparation for fusion with spectral information. Hence, the final result generated by the SpaM can be derived from

$$F_{\text{spa}} = \text{GELU}(\text{LN}(\text{linear}(T_{\text{out}}))). \tag{26}$$

C. Short- and Long-Range Spectral Dynamic Perception

Short-range spectral features facilitate the identification of local variations within individual spectral curves, and the long-range spectral correlation provides broader contextual data that aids in characterizing differences among spectral information of various samples. To exploit these two features, the SpeM is proposed. In this subsection, we detail the methodology adopted for achieving short- and long-range spectral perception by using Mamba. The structure of SpeM is shown in Fig. 4.

1) Multiscale Spectral Information Extraction: Before spectral grouping, we employ the MSpeFE to enhance the interaction among these groups and to bolster Mamba's capability for short-range spectral dynamic analysis. Specifically, we can formulate the MSpeFE module as

$$F_1^{\text{spe}} = \text{GELU}(\text{LN}(\text{Con}_{1\times1\times7}(F_s))) \tag{27}$$
$$F_2^{\text{spe}} = \text{GELU}(\text{LN}(\text{Con}_{1\times1\times3}(F_s))) \tag{28}$$
$$F_{ms}^{\text{spe}} = \text{Avgpool}_{H\times W}\left(F_1^{\text{spe}} + F_1^{\text{spe}} + F_s\right) \tag{29}$$

where F_1^{spe}, $F_2^{\text{spe}} \in \mathbb{R}^{H\times W\times c}$ denotes the multiscale spectral information, $\text{Avgpool}_{H\times W}(.)$ is the global spatial average pooling, and $\text{Con}_{1\times1\times3}(.)$ and $\text{Con}_{1\times1\times7}(.)$ are the 3-D convolution. To preserve the spectral sequence, we integrate multiscale spectral information using an element-wise sum.

2) Short-Range Spectral Grouping: We first define two parameters m and n. The number of channels of F_{ms}^{spe} is c, and the result output by the grouping module is $X \in \mathbb{R}^{g\times m}$. Therefore, the relationship among g, n, m, and c can be expressed by the following formula:

$$g = \left\lfloor \frac{c-(m-n)}{n} \right\rfloor + \left\lceil \frac{(c-(m-n)) \bmod n + m - n}{n} \right\rceil \tag{30}$$

where $\lfloor\cdot\rfloor$ and $\lceil\cdot\rceil$ are the floor function and the ceiling function, respectively. After the band grouping, the original spectral information is transformed from a c-dimensional sequence into a g-dimensional sequence ($g \ll c$), thereby reducing redundant spectral information. Furthermore, the sequential generation of the spectral sequence also preserves the directionality of spectral information and can be naturally combined with Mamba to model long-range spectral correlations.

3) Long-Range SDCC: The SDCC also employs the Mamba block to handle the features, and this process can be represented as

$$X_x, X_z = \text{chunk}(\text{linear}(\text{LN}(X))) \tag{31}$$
$$X_x^m = \text{S6}(\text{Silu}(\text{Conv}(X_x))) \tag{32}$$
$$X_{\text{out}} = \text{linear}\left(\text{LN}\left(X_x^m\right) \odot \text{Silu}(X_z)\right) + X \tag{33}$$

where X_x, X_z, $X_x^m \in \mathbb{R}^{g\times 2m}$ are the intermediate variables of SDCC, and X_{out} is the output of SDCC. Let us assume that the output of $\text{Silu}(\text{Conv}(X_x))$ is $X_x^s = [x_{1,x}, x_{2,x}, \ldots, x_{i,x}, \ldots, x_{g,x}]^T \in \mathbb{R}^{g\times 2m}$, $X_x^m = [x_{1,x}^m, x_{2,x}^m, \ldots, x_{i,x}^m, \ldots, x_{g,x}^m]^T$; then, the processing of the S6 for X_x^s can be expressed as

$$h_i = \overline{A}_i \odot h_{i-1} + B_i(\Delta_i \odot x_{i,x}) \tag{34}$$
$$x_{i,x}^m = C_i h_i + D \odot x_{i,x} \tag{35}$$

where $\overline{A} = \exp(\Delta A) \in \mathbb{R}^{d\times 2m}$ denotes the parameters for the control state mapping after discretization, and $h_i, h_{i-1} \in \mathbb{R}^{d\times 2m}$ are the states corresponding to different positions in the sequence. B, C, and Δ depend on the input X_x^s. Specifically, S6 utilizes $B = (X_x^s W_B)^{\text{T}}$, $C = X_x^s W_c$, and $\Delta = X_x^s W_r W_s$ to generate parameters $B \in \mathbb{R}^{d\times g}$, $C \in \mathbb{R}^{g\times d}$, and $\Delta \in \mathbb{R}^{g\times 2m}$, where $W_B, W_C \in \mathbb{R}^{2m\times d}$, $W_r \in \mathbb{R}^{2m\times C_0}$, and $W_s \in \mathbb{R}^{C_0\times 2m}$ represent mapping matrices. It can be seen from (34) and (35) that the correlation information of adjacent spectra is transmitted through the update and transfer of h, and the long-range dependencies are explored by the SDCC in a manner that conforms to the directional properties of spectral information.

Finally, we restore X_{out} into a 1-D sequence $F_{\text{spe}}^m \in \mathbb{R}^{1\times gm}$ and use a linear mapping to obtain the spectral weights at the corresponding positions. This can be expressed as

$$F_{\text{spe}} = \text{GELU}\left(\text{LN}\left(\text{linear}\left(F_{\text{spe}}^m\right)\right)\right). \tag{36}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES USED IN THE IP, PU, UH2013, AND SA DATASETS

IP dataset						PU dataset			UH dataset						SA dataset					
Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	29	25	10	470	503	1	548	6304	1	198	1053	10	191	1036	1	31	1978	10	33	3245
2	762	675	11	1414	1065	2	540	18146	2	190	1064	11	181	1054	2	38	3688	11	11	1057
3	435	404	12	328	282	3	392	1815	3	192	505	12	192	1041	3	20	1956	12	20	1907
4	146	99	13	132	80	4	524	2912	4	188	1056	13	184	285	4	14	1380	13	10	906
5	232	274	14	728	545	5	265	1113	5	186	1056	14	181	247	5	27	2651	14	11	1059
6	394	354	15	291	99	6	532	4572	6	182	143	15	187	473	6	40	3919	15	73	7195
7	16	12	16	57	44	7	375	981	7	196	1072				7	36	3543	16	19	1788
8	235	250				8	514	3364	8	191	1053				8	113	11158			
9	10	10				9	231	795	9	193	1059				9	63	6140			

D. Objective Function

To optimize the model's parameters, we implement a joint-constrained method that balances sample distribution and minimizes the correlation among spectral weights. First, we employ the focal loss to compute class-level losses across global samples, which strengthens the model's ability to address sample imbalance issues, formulated as

$$t_{kij} = \begin{cases} 1, & \text{if } k = \overline{y}_{ij} \\ 0, & \text{if } k \neq \overline{y}_{ij} \end{cases} \tag{37}$$

$$L_f = -\frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{k=1}^{K} (\alpha_k \times (1 - p_k)^{\gamma} \times \log(p_k) \times t_{kij}) \tag{38}$$

where $\overline{y}_{ij}$ is the ground truth, α is a class-balancing parameter, and γ denotes a focusing parameter. In addition, our MambaLG employs the L1-norm to implement deep supervision on the output of SpeM, ensuring that the SpeM achieves global spectral perception. It can be formulated as

$$\text{count} = \sum_{i=1}^{c} 1_{w_i > t} \tag{39}$$

$$L_s = \frac{\lambda}{K} \left\| F_{\text{spe}} \right\|_1 + |T - \text{count}| \tag{40}$$

where λ is a balancing parameter. T ensures that not all elements in F_{spe} are set to zero, but at least T data points should be greater than t. Thus, the overall loss is given by

$$L = L_f + L_s \tag{41}$$

where η is a balancing parameter.

IV. EXPERIMENTS

In this section, we conduct extensive experiments to evaluate the advantages of our MambaLG and assess the performance of its various modules. These experiments are carried out on four well-known HSI datasets, which differ in terms of spatial resolution and spectral coverage. The network architectures are implemented using the PyTorch framework using a workstation with an NVIDIA GeForce GTX 4090 GPU.

A. Dataset Description

Three classic spatially disjoint datasets are employed, namely the Indian Pines (IP) dataset, the Pavia University (PU) dataset, and the Houston University 2013 (HU) dataset. Additionally, to evaluate the performance of the model in the presence of limited samples, this section also utilizes the Salinas (SA) dataset. The allocation of training and testing sets on these four datasets in the subsequent experiments is shown in Table I. The spatial distribution of the training and testing datasets for the IP dataset follows the same settings described in [32], while those for the PU and UH datasets follow the settings of Spectralformer.[1] Detailed information on the four datasets is given below.

1) The IP dataset, acquired by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS), consists of 145 × 145 pixels, each with a resolution of 20 m, and covers 200 spectral bands in the 400–2500-nm range. The dataset contains 10 400 labeled pixels across 16 ground-truth classes.
2) The PU dataset was acquired using the reflective optics spectrographic imaging system (ROSIS) sensor over Northern Italy, featuring a 610 × 340 pixel area with nine classes, a spatial resolution of 1.3 m, and 103 spectral bands ranging from 430 to 860 nm.
3) The UH dataset, captured by an ITRES CASI-1500 sensor, comprises 349 × 1905 pixels with 144 spectral bands in the 380–1050-nm range. The dataset has a spatial resolution of 1.5 m and contains 15 distinct classes.
4) The SA dataset, depicting the Salinas Valley in California, USA, and acquired by AVIRIS, is a 512 × 217 pixel image with a spatial resolution of 3.7 m, 16 classes, and includes 204 spectral bands from 360 to 2500 nm.[2] A subset of 1% of the samples per class is selected for network training, with the remaining samples allocated for testing. Note that the spatial distribution of the training and testing data for the SA dataset is not given, as the training data points are randomly selected.

[1] https://github.com/danfenghong/IEEE_TGRS_SpectralFormer

[2] https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

TABLE II

OAs, AAs (%), and Kappa Achieved by Different Methods on the IP Dataset. The Best Result in Each Row Is Bolded

Class	CNN, RNN, GCN, FCN-Based				Transformer-Based				MambaLG
	A2S2K	ASSMN	FDGC	SSDGL	HiT	SSTN	SSFTT	SPRLT	
1	45.24	**100.00**	48.69	**100.00**	76.64	67.33	70.25	87.14	48.00
2	90.14	87.41	71.21	**90.52**	75.24	72.72	69.86	84.23	85.78
3	**95.20**	76.90	82.29	81.44	76.97	93.69	79.35	88.38	93.07
4	67.99	52.86	81.73	**84.85**	57.40	83.86	64.11	79.77	52.53
5	94.32	42.58	87.77	91.61	88.00	95.60	91.15	**96.64**	73.75
6	96.19	92.56	87.92	98.31	97.67	96.55	97.60	98.55	**99.15**
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	97.12	94.67	99.84	**100.00**	96.68	99.61	99.84	99.76	**100.00**
9	75.56	80.00	71.11	**100.00**	93.50	67.61	54.73	**100.00**	0.00
10	87.99	58.25	72.62	60.03	82.72	86.40	84.43	87.38	**93.24**
11	87.68	91.74	86.07	85.63	77.60	95.71	84.01	87.06	**95.87**
12	69.74	90.19	53.16	89.01	51.79	76.58	67.54	67.24	**92.55**
13	89.55	97.50	63.06	82.50	97.33	81.93	71.10	90.56	**98.75**
14	98.68	98.35	94.71	**99.63**	86.08	97.98	98.94	95.22	98.89
15	88.95	47.81	44.27	10.10	87.47	84.50	46.97	**90.31**	77.77
16	63.02	81.06	27.57	84.09	68.05	62.17	39.50	73.03	**86.36**
OA(%)	88.84	82.53	78.30	85.52	79.84	86.65	81.41	87.84	**91.38**
AA(%)	77.96	74.49	67.00	78.61	75.82	78.88	69.96	**82.82**	74.71
kappa	0.8732	0.7993	0.7542	0.8353	0.7702	0.8487	0.7889	0.8617	**0.9019**

TABLE III

OAs, AAs (%), and Kappa Achieved by Different Methods on the PU Dataset. The Best Result in Each Row Is Bolded

Class	CNN, RNN, GCN, FCN-Based				Transformer-Based				MambaLG
	A2S2K	ASSMN	FDGC	SSDGL	HiT	SSTN	SSFTT	SPRLT	
1	98.96	84.84	87.14	85.23	96.64	96.86	95.82	**99.42**	89.94
2	**99.58**	92.38	98.27	89.59	93.55	93.46	98.00	98.87	95.88
3	90.29	53.04	92.44	64.08	83.22	94.51	85.91	**95.42**	87.93
4	94.97	**98.93**	54.22	98.11	87.02	91.96	87.25	86.18	98.90
5	99.50	99.75	92.62	99.19	98.50	99.64	99.24	98.96	**100.00**
6	48.65	49.73	49.74	65.90	81.61	94.60	49.29	92.04	**100.00**
7	**97.16**	81.41	96.01	80.22	81.90	82.00	90.51	93.54	92.76
8	77.60	97.19	69.61	97.47	69.67	84.92	83.25	79.84	**99.38**
9	98.07	99.02	77.72	**99.87**	95.52	89.48	84.74	95.37	98.36
OA(%)	84.40	85.48	77.77	86.56	88.87	92.56	83.68	94.74	**95.66**
AA(%)	89.42	84.03	79.75	86.63	87.51	91.93	86.00	93.29	**95.91**
kappa	0.8004	0.8027	0.7169	0.8197	0.8499	0.8985	0.7900	0.9295	**0.9419**

TABLE IV

OAs, AAs (%), and Kappa Achieved by Different Methods on the UH Dataset. The Best Result in Each Row Is Bolded

Class	CNN, RNN, GCN, FCN-Based				Transformer-Based				MambaLG
	A2S2K	ASSMN	FDGC	SSDGL	HiT	SSTN	SSFTT	SPRLT	
1	98.79	77.97	99.63	81.39	98.73	98.54	99.77	**100.00**	82.05
2	97.80	70.68	99.23	92.76	86.85	72.36	93.37	87.50	**100.00**
3	99.56	36.83	90.31	94.65	90.93	92.48	**100.00**	82.81	98.02
4	91.49	80.30	93.74	98.11	93.84	98.29	84.83	96.51	**99.81**
5	99.27	96.78	**100.00**	99.91	93.77	96.18	99.98	97.69	**100.00**
6	57.37	83.22	50.37	95.80	80.98	90.91	58.55	54.58	**97.20**
7	95.07	81.34	78.89	90.21	86.64	**99.46**	94.44	94.98	91.04
8	80.52	62.11	63.71	74.26	87.22	80.54	88.47	**89.43**	86.23
9	86.11	63.93	84.44	76.30	84.80	92.50	87.11	**95.42**	87.54
10	79.18	62.74	88.53	80.12	89.56	83.23	**98.30**	91.60	74.03
11	80.73	73.81	79.68	65.84	78.66	53.35	73.95	75.22	**90.04**
12	93.04	58.41	84.16	80.59	93.30	86.67	**95.64**	91.31	91.93
13	95.27	37.19	63.25	84.21	74.71	**98.15**	66.94	93.97	91.23
14	93.91	**100.00**	99.60	98.79	96.69	**100.00**	74.42	**100.00**	**100.00**
15	97.29	21.35	97.19	83.09	92.68	95.63	97.16	95.17	**100.00**
OA(%)	89.70	69.21	85.45	84.83	88.79	84.01	87.10	90.42	**91.30**
AA(%)	89.69	67.11	84.85	86.40	88.63	89.22	87.53	89.75	**92.61**
kappa	0.8882	0.6655	0.8423	0.8354	0.8783	0.8264	0.8604	0.8961	**0.9057**

TABLE V

OAs, AAs (%), and Kappa Achieved by Different Methods on the SA Dataset. The Best Result in Each Row Is Bolded

Class	CNN, RNN, GCN, FCN-Based				Transformer-Based				MambaLG
	A2S2K	ASSMN	FDGC	SSDGL	HiT	SSTN	SSFTT	SPRLT	
1	99.80	**100.00**	**100.00**	90.79	99.80	99.64	**100.00**	**100.00**	**100.00**
2	99.97	**100.00**	**100.00**	99.73	**100.00**	98.87	**100.00**	**100.00**	**100.00**
3	98.44	99.28	**100.00**	**100.00**	97.65	93.80	99.64	99.69	**100.00**
4	97.94	**100.00**	99.13	99.86	98.79	88.21	97.56	95.82	99.71
5	**99.81**	99.25	99.36	99.62	96.75	99.87	97.75	**99.81**	98.64
6	**100.00**	**100.00**	99.90	**100.00**	**100.00**	**100.00**	99.95	**100.00**	**100.00**
7	99.19	97.71	**100.00**	**100.00**	**100.00**	99.60	**100.00**	99.97	**100.00**
8	95.65	45.44	99.47	99.87	87.28	97.34	96.61	84.71	**99.78**
9	99.53	**100.00**	**100.00**	**100.00**	99.29	99.92	99.92	99.21	**100.00**
10	99.81	97.57	99.52	**99.87**	94.40	98.91	99.38	97.40	99.14
11	95.71	97.26	90.64	**100.00**	86.46	94.44	96.61	96.06	94.13
12	96.12	98.79	97.84	99.48	99.63	99.53	**99.95**	99.69	99.69
13	99.66	**100.00**	94.44	**100.00**	99.12	97.52	99.89	99.89	**100.00**
14	97.04	97.64	96.78	**100.00**	97.14	98.02	99.43	99.62	98.68
15	91.70	97.47	99.87	99.81	67.68	59.80	92.81	88.41	**99.89**
16	99.00	87.14	**100.00**	91.55	95.47	99.10	**100.00**	**100.00**	99.33
OA(%)	97.39	87.36	99.35	99.26	91.18	89.83	97.99	94.70	**99.63**
AA(%)	98.08	94.85	98.56	98.78	94.97	95.28	98.72	97.52	**99.31**
kappa	0.9710	86.05	0.9927	0.9917	0.9021	0.8876	0.9777	0.9409	**0.9959**

B. Experimental Setup

1) Comparison With State-of-the-Art Networks: To assess the performance of our MambaLG, we select some classification models utilizing diverse DL architectures. These methods include the CNN-based attention-based adaptive spectral–spatial kernel improved residual network (A2S2K) [13], the RNN-based adaptive spectral–spatial multiscale network (ASSMN) [36], the graph convolutional network (GCN)-based fast dynamic GCN (FDGC) [37], and several transformer-based models such as the HSI transformer (HiT) [38], spectral–spatial transformer network (SSTN) [34], spectral–spatial feature tokenization transformer (SSFTT) [39], and local transformer with spatial partition restore network (SPRLT) [32]. The CNN-based, GCN-based, and transformer-based methods are patch-based. In addition, the spectral–spatial-dependent global learning framework (SSDGL) [40] is selected to evaluate the advantages of our MambaLG over the global learning framework.

2) Model Training and Testing: We train the network for 1000 epochs on both the IP and PU datasets, and for 500 epochs on both the UH and SA datasets, adopting the Adam optimizer [41] with a learning rate of 0.0005. For a fair comparison, we use the optimal parameters reported in each publication to train the algorithms. To quantitatively assess the performance of the models, we employ four metrics, including overall accuracy (OA), average accuracy (AA), kappa, and class-specific classification accuracy (CA).

C. Comparative Experimental Results and Analysis

1) Quantitative Comparison: The quantitative results of MambaLG compared with other algorithms on four datasets are detailed in Tables II–V. On the IP dataset, A2S2K achieves the best OA and kappa among other comparison algorithms, while SPRLT obtains the best AA. In contrast, MambaLG improves on A2S2K's OA and kappa by 2.54% and 0.0287%, respectively. Although MambaLG's AA does not stand out, it achieves the highest number of optimal CA instances. On the PU and UH datasets, Transformer-based methods outperform the other comparison network-based methods, indicating that the extraction of long-range information can effectively enhance the model performance. On the SA dataset, FDGC and SSDGL (which utilize GCN and FCN structures) achieve better results, underscoring the importance of spatial structural correlation and global semantic mining. To capitalize on these two types of information, MambaLG incorporates LSCE and GSCE, thus also achieving the best quantitative results on the SA dataset. Notably, both A2S2K and SSDGL belong to the dual-branch methods but feature coupling, resulting in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

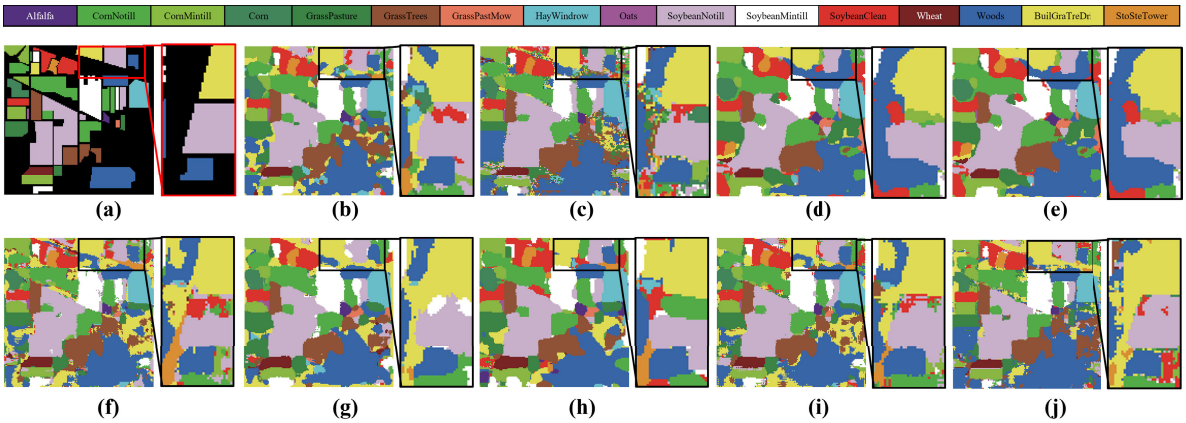


Fig. 5. Visual quality comparison on the IP dataset, which shows the (a) ground truth, and the classification maps corresponding to (b) A2S2K, (c) ASSMN, (d) FDGC, (e) SSDGL, (f) HiT, (g) SSTN, (h) SSFTT, (i) SPRLT, and (j) MambaLG, respectively.

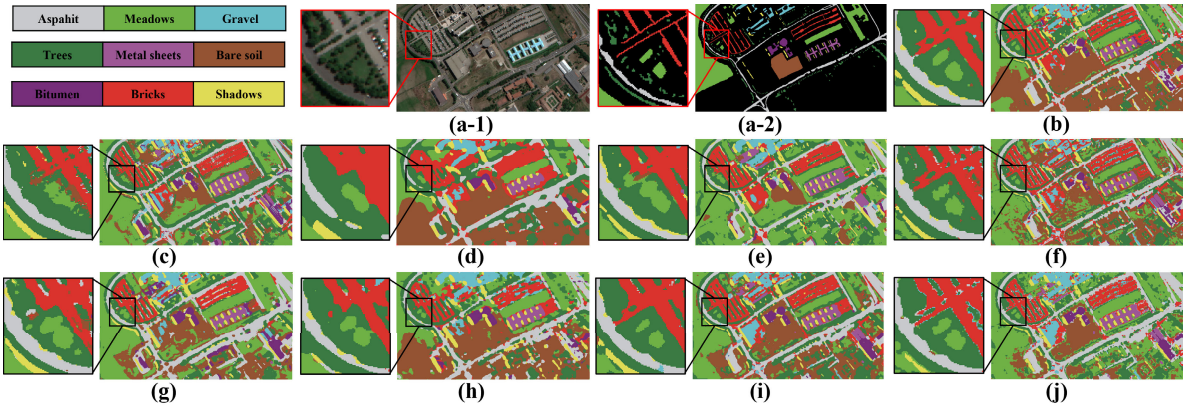


Fig. 6. Visual quality comparison on the PU dataset, which shows the (a-1) pseudocolor image, (a-2) ground truth, and the classification maps corresponding to (b) A2S2K, (c) ASSMN, (d) FDGC, (e) SSDGL, (f) HiT, (g) SSTN, (h) SSFTT, (i) SPRLT, and (j) MambaLG, respectively.

inferior performance compared to MambaLG. In summary, the MambaLG that we proposed achieves the best quantitative results. These results reveal that the parallel spectral–spatial decoupling approach, along with the coupling approach featuring functional interaction, can effectively enhance the model's robustness. Furthermore, considering local and global information has a significant impact on enhancing the model's performance, which validates the value of our improvements to some extent.

2) Qualitative Comparison: Figs. 5–8 display the classification maps of various algorithms on four datasets. On the IP dataset, MambaLG exhibits the strongest edge preservation and more prominent details. It also showcases reduced noise on the PU dataset. In the UH dataset, the road segmentation results on the right side reveal that MambaLG, A2S2K, ASSMN, and HiT effectively maintain the road's shape. Conversely, the classification result of the outdoor sports field on the left side shows that MambaLG exhibits the lowest misclassification rate and preserves a more complete ground feature structure. On the SA dataset, the classification maps of SSGDL and MambaLG clearly delineate the road section, indicating the global learning framework's robust recognition capability for challenging ground features. Furthermore, SSGDL experiences significant misclassifications at category edges, whereas MambaLG effectively mitigates this issue, leading to improved results. In general, patch-based methods tend to be over-smooth, whereas the global learning framework SSGDL (which employs encoding and decoding) minimizes noise influence but may result in the loss of local structures. In contrast, MambaLG, grounded in Mamba, achieves global information mining, enhances the model's recognition of ground features that are easy to confuse, and employs a multiscale strategy to safeguard the integrity of local details.

3) Model Parameters and Efficiency Comparison: Table VI presents the time required for each algorithm to conduct a single training epoch and perform full-image inference on various datasets, as well as the number of training parameters. It is evident that SSDGL and MambaLG, which utilize a global learning framework, significantly outperform other methods in terms of image inference speed. Despite the need to process the entire dataset during training, these global models still exhibit superior training speed. Regarding model parameters, SSTN and SSFTT (which operate on local patches) require the lowest number of parameters. While their parameter counts are only $0.1\times$ and $0.79\times$ those of MambaLG on the PU dataset, MambaLG's inference times require only 0.004% and 0.01% of the time required by these two methods, respectively. Moreover, MambaLG has a substantially lower parameter count compared to SSDGL, indicating that Mamba effectively minimizes the number of parameters required by global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.



Fig. 7. Visual quality comparison on the UH dataset, which shows the (a) pseudocolor image, and the classification maps corresponding to (b) A2S2K, (c) ASSMN, (d) FDGC, (e) SSDGL, (f) HiT, (g) SSTN, (h) SSFTT, (i) SPRLT, and (j) MambaLG, respectively.

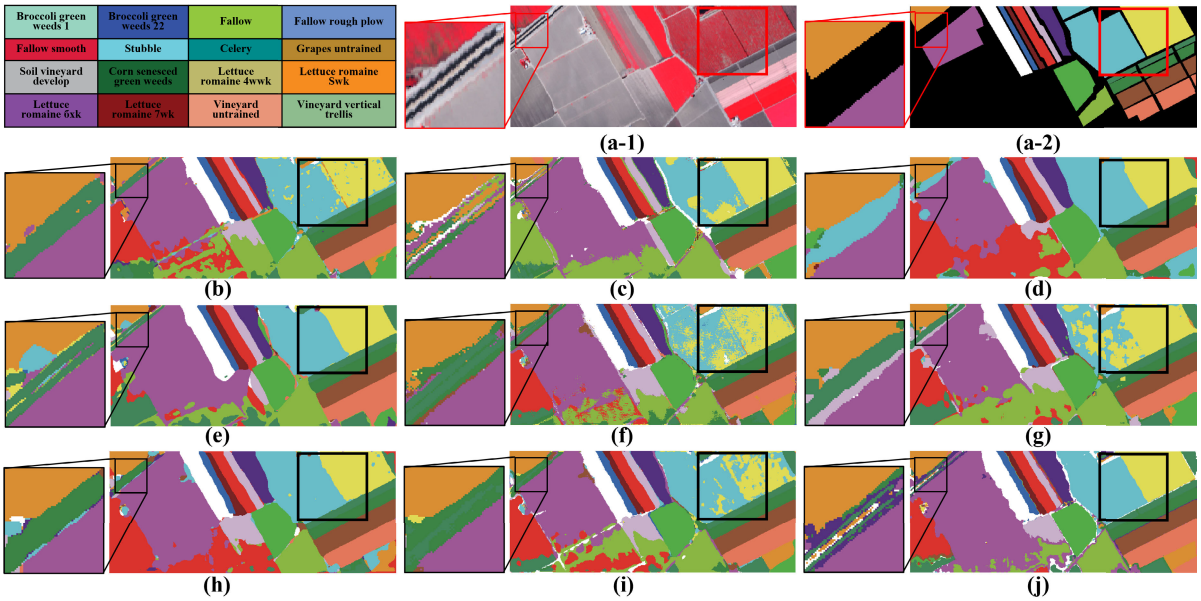


Fig. 8. Visual quality comparison on the SA datasets, which shows the (a-1) pseudocolor image, (a-2) ground truth, and the classification maps corresponding to (b) A2S2K, (c) ASSMN, (d) FDGC, (e) SSDGL, (f) HiT, (g) SSTN, (h) SSFTT, (i) SPRLT, and (j) MambaLG, respectively.

TABLE VI

TRAINING TIME, TEST TIME, AND NUMBER OF PARAMETERS REQUIRED BY DIFFERENT METHODS WHEN PROCESSING THE IP, PU, UH, AND SA DATASETS. THE BEST RESULT IN EACH ROW IS BOLDED AND THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	IP dataset			PU dataset			UH dataset			SA dataset		
	Train	Test	Params	Train	Test	Params	Train	Test	Params	Train	Test	Params
A2S2K	9.31	16.41	0.371M	5.84	111.08	0.221M	4.368	35.53	0.283M	0.892	14.57	0.377M
ASSMN	18.24	21.83	2.869M	12.89	56.54	2.696M	8.76	37.00	2.774M	1.82	740.19	2.876M
FDGC	0.873	1.93	2.384M	0.610	15.45	1.921M	0.403	4.67	2.318M	0.086	4.72	2.384M
SSDGL	**0.126**	**0.126**	3.504M	**0.171**	**0.171**	1.718M	**0.245**	**0.245**	1.802M	0.113	**0.113**	3.515M
HiT	5.996	17.11	59.286M	3.877	159.03	54.504M	2.889	45.24	56.141M	0.626	27.96	59.549M
SSTN	6.655	16.84	**0.021M**	4.481	133.34	**0.016M**	3.057	28.40	**0.018M**	0.606	14.44	**0.021M**
SSFTT	0.769	7.78	0.153M	0.506	59.56	0.153M	0.343	16.90	0.153M	**0.079**	2.74	0.153M
SPRLT	1.944	9.20	0.846M	1.257	65.76	0.833M	0.969	20.01	0.839M	0.207	11.24	0.847M
MambaLG	0.251	0.251	0.193M	0.598	0.598	0.186M	1.641	1.641	0.189M	0.543	0.543	0.194M

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

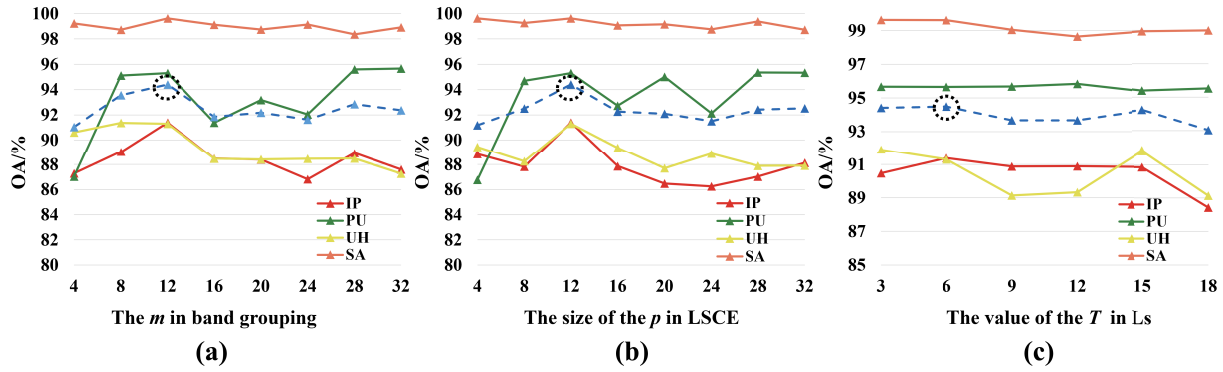


Fig. 9. Effect of different values of MambaLG parameters: (a) m, (b) p, and (c) T on the four datasets.

framework models. In summary, our MambaLG is one of the top performers regarding training speed, inference speed, and parameter count.

V. Discussion

In this section, we first examine the impact of hyperparameters on model performance to indirectly demonstrate the value of the modules proposed. Second, extensive ablation studies are conducted to concretely verify the performance of each module in HSI classification.

A. Hyperparameter Analysis for MambaLG

In SpaM, the LSCE is designed to model local spatial correlations, with the spatial patch size p being a critical parameter. In SpeM, both the number of spectral groups m and the value of T within L_s influence the sparsity of F_{spe}. This subsection analyzes the impact of these parameters on model performance. Fig. 9(a) illustrates the OA across four datasets as m varies within the range $\{4, 8, 12, 16, 20, 24, 28, 32\}$. The accuracy initially increases with increasing values of m and then declines, suggesting that, while global spectral perception is advantageous, excessive weight sparsity may lead to information loss. This trend is consistent with basic cognitive principles and corroborates the efficacy of spectral grouping. Fig. 9(b) depicts the model accuracy changes with the variation of p, ranging in our experiments from $\{4, 8, 12, 16, 20, 24, 28, 32\}$. The model demonstrates enhanced robustness and generalization when the spatial token size is 12. However, large spatial tokens hurt model performance on the IP dataset with low resolution and the UH dataset with complex structures. The poor results are likely due to the introduction of excessive information through larger spatial correlation modeling, which can hinder the extraction of key spatial features. The performance variations as the parameter T takes values in the range $\{3, 6, 9, 12, 15, 18\}$ are presented in Fig. 9(c). This figure demonstrates that the model achieves optimal performance with a parameter value of 6, confirming that appropriate spectral sparsity positively influences the model's robustness. In conclusion, these results validate the effectiveness of SpaM, SpeM, and spectral constraints and indirectly demonstrate the importance of local spatial information details and spectral sparsity for HSI classification.

TABLE VII

OAs (%), AAs (%), and Kappa Achieved by Different Configurations for the IP and PU Datasets. The Best Result in Each Row Is Bolded

The modules of MambaLG					IP dataset			PU dataset		
GSCE	MSFE	LSCE	SpeM	L1	OA	AA	Kappa	OA	AA	Kappa
✓					88.6	81.48	0.8703	85.36	91.34	0.8117
✓	✓				89.38	83.12	0.8790	93.56	95.53	0.9148
✓	✓	✓			91.06	84.28	0.8983	94.56	95.76	0.9277
✓	✓	✓	✓		91.19	**84.53**	0.8998	94.63	95.88	0.9286
✓	✓	✓	✓	✓	**91.38**	74.71	**0.9019**	**95.66**	**95.91**	**0.9419**

B. Quantitative Analysis of Module Ablation

We utilize the GSCE as the base model and conduct experiments by incrementally adding modules to analyze their impact on the model's performance. The experimental results on the IP and PU datasets are presented in Table VII. As shown in Table VII, we find that all modules effectively enhance model performance. Specifically, the MSFE significantly improves the model's performance on the PU dataset, indicating that multiscale spatial strategies can mitigate the destruction of spatial structures caused by the sequential operations in Mamba. The LSCE increases the model's OA by 1.68% on the IP dataset and by 1% on the PU dataset, underscoring the importance of integrating local and global spatial information for improving model performance.

In the spectral feature extraction, both SpeM and the L1-norm have been found to enhance model performance. SpeM's effectiveness corroborates the widely accepted notion that the joint utilization of spatial and spectral information significantly contributes to model performance, thereby indirectly validating the scientific rationale of MambaLG. Notably, the application of the L1-norm has led to a marked improvement in the model's accuracy on the PU dataset, increasing from 94.63% to 95.66%. This result further underscores the significance of imposing prior constraints on the spectral branch.

C. Module for Category Separability

To more intuitively demonstrate the ability of each module to enhance the model's capability to recognize different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

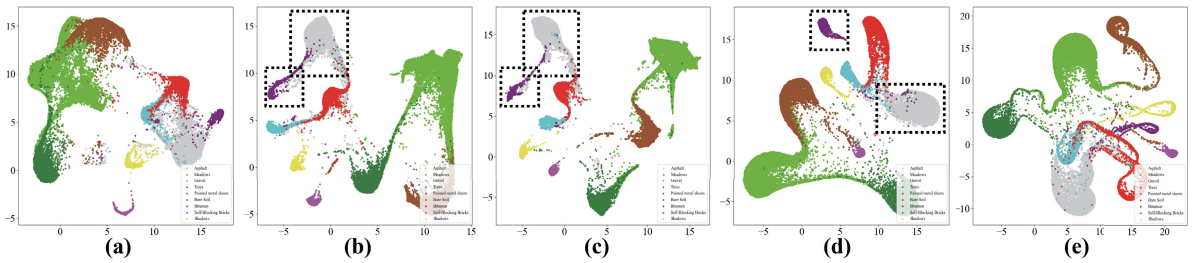


Fig. 10. Visualization of T-SNE results on the PU test samples by different methods, including (a) GSCE, (b) GSCE + MSFE, (c) GSCE + MSFE + LSCE, (d) GSCE + MSFE + LSCE + SpeM, and (e) GSCE + MSFE + LSCE + SpeM + L1.

TABLE VIII

OAs (%), AAs (%), AND KAPPA ACHIEVED BY MAMBALG WITH DIFFERENT FUSION METHODS FOR THE PU AND SA DATASETS. THE BEST RESULT IN EACH ROW IS BOLDED. $\oplus$ AND $\odot$ REPRESENT MULTIPLICATION AND ADDITION, RESPECTIVELY

The fusion method	PU dataset			SA dataset		
	OA	AA	Kappa	OA	AA	Kappa
pixel-wise $\oplus$	91.36	95.14	0.8868	93.31	95.55	0.9256
Weighted $\oplus$	90.38	94.93	0.8745	93.12	95.42	0.9235
Gated fusion $\oplus$	89.96	95.22	0.8694	93.33	95.78	0.9259
Cascading+Conv	90.41	94.91	0.8749	90.92	94.49	0.8991
pixel-wise $\odot$	95.44	**95.99**	0.9389	95.58	**96.09**	0.9509
Channel-wise $\odot$	**95.66**	95.91	**0.9419**	**95.86**	95.85	**0.9539**

categories, we visualize the outputs for different modules selected by MambaLG, as shown in Fig. 10. Notably, the visualization focuses on F_{fusion} rather than F_{out}, effectively mitigating the influence of the output layer and aiding in the analysis of the performance of different spatial and spectral modules. Fig. 10(a) shows that the original feature separability is poor, with minimal variance between classes. As more modules are incorporated, the feature distributions for different classes become increasingly distinct. Particularly, for classes with similar compositions (such as Asphalt and Bitumen), the fusion of spectral and spatial information significantly enhances the differences between classes. In addition, comparison of Fig. 10(a) with Fig. 10(c) shows that the addition of MSFE and LSCE significantly enhances the performance of the method with only GSCE, demonstrating the importance of preserving local spatial structure for the extraction of global spatial context.

D. Comparison of Different Fusion Methods

Table VIII displays the performance of MambaLG on the PU and SA datasets when different postprocessing fusion methods are employed. Note that to better represent the model differences on the SA dataset, this experiment selects only 15 samples per class for training. As shown in Table VIII, on the PU dataset, pixel-wise addition outperforms weighted addition and gated fusion addition. This suggests (to a certain extent) that more complex fusion methods do not necessarily yield better results. The fusion method, which combines cascading and convolution, demonstrates comparable performance to addition fusion on the PU dataset; however, it exhibits inferior performance on the SA dataset. This can be attributed to the model's tendency to rely excessively on the fusion architecture, resulting in an uncontrollable (and consequently weaker) capacity for generalization. Although pixel-wise point multiplication yields comparable outcomes to those of channel-wise point multiplication, the former requires the processing of a greater quantity of data, exerting significant pressure on the hardware. In contrast, channel-wise point multiplication can achieve superior results with less computational effort, further underscoring the advantage of our channel-level point multiplication fusion method within the local-to-global framework.

In summary, the experimental results presented in this section confirm the efficacy of the innovations within MambaLG for the extraction of both local and global information, and for the functional integration of spatial and spectral features. Notably, our findings indicate that spectral sparsity constraints exert a substantial influence on model performance. Furthermore, we have observed that simple functional interaction-based data fusion approaches may outperform more intricate methods. These observations underscore the significant potential for future research aimed at refining methodologies based on the inherent characteristics of HSIs.

VI. CONCLUSION

In this article, we propose a new Mamba-based method (MambaLG) to extract local to global spatial and spectral features for HSI classification. The model decouples spatial and spectral information through a parallel feature extraction strategy and achieves the fusion of spectral and spatial information through functional-level interaction. Extensive experiments on widely used datasets with different spatial resolutions and spectral coverage demonstrate that our MambaLG outperforms other state-of-the-art algorithms and effectively showcases the performance of each module. In particular, the spatial feature extraction strategy realizes effective protection of the local spatial detail while modeling the global spatial semantics. In the spectral feature extraction strategy, spectral grouping and spectral dynamic clustering are combined to generate the channel weights that effectively perceive differences in terms of short- and long-range spectral importance. Adding spectral sparsity constraints to the loss function can effectively enhance the generalization and interpretability of the model. In the future, we will study extraction methods that are more in line with the integrated nature of spatial and spectral information of HSIs based on a local-to-global framework.

ACKNOWLEDGMENT

The numerical calculations in this study were carried out on a theorized supercomputer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] S. Bhadra, V. Sagan, S. Sarkar, M. Braud, T. C. Mockler, and A. L. Eveland, "PROSAIL-Net: A transfer learning-based dual stream neural network to estimate leaf chlorophyll and leaf angle of crops from UAV hyperspectral images," *ISPRS J. Photogramm. Remote Sens.*, vol. 210, pp. 1–24, Apr. 2024.

[2] J. Guo, D. Hong, and X. X. Zhu, "High-resolution satellite images reveal the prevalent positive indirect impact of urbanization on urban tree canopy coverage in South America," *Landscape Urban Planning*, vol. 247, Jul. 2024, Art. no. 105076.

[3] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2018.

[4] M. A. Ali, X. Lyu, M. S. Ersan, and F. Xiao, "Critical evaluation of hyperspectral imaging technology for detection and quantification of microplastics in soil," *J. Hazardous Mater.*, vol. 476, Sep. 2024, Art. no. 135041.

[5] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[6] B. Xi, Y. Zhang, J. Li, Y. Li, Z. Li, and J. Chanussot, "CTF-SSCL: CNN-transformer for few-shot hyperspectral image classification assisted by semisupervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532617.

[7] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 4, 2024, doi: 10.1109/TNNLS.2024.3401589.

[8] V. Kumar, R. S. Singh, M. Rambabu, and Y. Dua, "Deep learning for hyperspectral image classification: A survey," *Comput. Sci. Rev.*, vol. 53, Aug. 2024, Art. no. 100658.

[9] J. Deng et al., "RustQNet: Multimodal deep learning for quantitative inversion of wheat stripe rust disease index," *Comput. Electron. Agricult.*, vol. 225, Oct. 2024, Art. no. 109245.

[10] D. Hong, C. Li, B. Zhang, N. Yokoya, J. A. Benediktsson, and J. Chanussot, "Multimodal artificial intelligence foundation models: Unleashing the power of remote sensing big data in Earth observation," *Innov. Geosci.*, vol. 2, no. 1, 2024, Art. no. 100055.

[11] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[12] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[13] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[14] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.

[15] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5612–5626, Aug. 2020.

[16] J. Bai, W. Shi, Z. Xiao, T. A. A. Ali, F. Ye, and L. Jiao, "Achieving better category separability for hyperspectral image classification: A spatial–spectral approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9621–9635, Jul. 2024.

[17] X. Hu, X. Wang, Y. Zhong, and L. Zhang, "S^{3}ANet: Spectral–spatial-scale attention network for end-to-end precise crop classification based on UAV-borne H2 imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 147–163, Jan. 2022.

[18] Z. Xie, J. Hu, X. Kang, P. Duan, and S. Li, "Multilayer global spectral–spatial attention network for wetland hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518913.

[19] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[20] K. Cao et al., "Shuffle mamba: State space models with random shuffle for multi-modal image fusion," 2024, *arXiv:2409.01728*.

[21] L. Wang, D. Li, S. Dong, X. Meng, X. Zhang, and D. Hong, "PyramidMamba: Rethinking pyramid feature fusion with selective space state model for semantic segmentation of remote sensing imagery," 2024, *arXiv:2406.10828*.

[22] A. Gu, T. Dao, S. Ermon, A. Rudra, and C. Ré, "HiPPO: Recurrent memory with optimal polynomial projections," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2020, pp. 1474–1487.

[23] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2022, *arXiv:2111.00396*.

[24] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[25] C. Yang et al., "A barking emotion recognition method based on mamba and synchrosqueezing short-time Fourier transform," *Expert Syst. Appl.*, vol. 258, Dec. 2024, Art. no. 125213.

[26] H. Guo, J. Li, T. Dai, Z. Ouyang, X. Ren, and S. Xia, "MambaIR: A simple baseline for image restoration with state-space model," in *Proc. ECCV*, Sep. 2024, pp. 222–241.

[27] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] C. Wang, J. Huang, M. Lv, H. Du, Y. Wu, and R. Qin, "A local enhanced mamba network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, Sep. 2024, Art. no. 104092.

[29] L. Huang, Y. Chen, and X. He, "Spectral–spatial mamba for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 13, p. 2449, Jul. 2024.

[30] W. Zhou, S.-I. Kamata, H. Wang, M.-S. Wong, Huiying, and Hou, "Mamba-in-mamba: Centralized mamba-cross-scan in tokenized mamba model for hyperspectral image classification," 2024, *arXiv:2405.12003*.

[31] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[32] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.

[33] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[34] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715.

[35] X. Li, D. Hong, and J. Chanussot, "S2MAE: A spatial–spectral pretraining foundation model for spectral remote sensing data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27696–27705.

[36] D. Wang, B. Du, L. Zhang, and Y. Xu, "Adaptive spectral–spatial multiscale contextual feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2461–2477, Mar. 2021.

[37] Q. Liu, Y. Dong, Y. Zhang, and H. Luo, "A fast dynamic graph convolutional network and CNN parallel network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530215.

[38] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.

[39] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[40] Q. Zhu et al., "A spectral–spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11709–11723, Nov. 2022.

[41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.



Zhaojie Pan (Student Member, IEEE) received the B.S. and M.Sc. degrees from China University of Petroleum (East China), Qingdao, China, in 2021 and 2024, respectively. He is currently pursuing the Ph.D. degree with the School of Mathematics, Southeast University, Nanjing, China.

He is also a joint Ph.D. student with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include hyperspectral image classification and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.



Chenyu Li (Student Member, IEEE) received the B.S. and M.S. degrees from the School of Transportation, Southeast University, Nanjing, China, in 2018 and 2021, respectively, where she is currently pursuing the Ph.D. degree from the Department of Mathematics.

She is also a joint Ph.D. Student with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. Her research interests include interpretable artificial intelligence, big Earth data forecasting, foundation models, and hyperspectral imaging.



Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees from the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain, in 1999 and 2002, respectively.

He is a Full Professor and the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura. His research interests comprise hyperspectral data processing and parallel computing of remote sensing data. He has authored 850 publications in this field.

Prof. Plaza is a fellow of AAIA and Academia Europaea. He was a recipient of the recognition of Best Reviewers of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and the recognition of Best Reviewers of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he served as an Associate Editor from 2007 to 2012. He was a recipient of the Best Column Award of the *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of the JSTARS journal, and the Most Highly Cited Paper from 2005 to 2010 in the *Journal of Parallel and Distributed Computing*. He served as the Director of Education Activities for the IEEE GRSS from 2011 to 2012, and the President of the Spanish Chapter of IEEE GRSS from 2012 to 2016. He is currently serving as the Chair of the Publications Awards Committee of IEEE GRSS and the Fellow Evaluations Committee of IEEE GRSS. He served as the Editor-in-Chief of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING journal for five years from 2013 to 2017 and as the Editor-in-Chief of IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS from 2019 to 2020. He has been included in the 2018–2023 Highly Cited Researchers List (Clarivate Analytics). He was also an Associate Editor of IEEE ACCESS (receiving the recognition of Outstanding Associate Editor for the journal in 2017). Additional information: http://sites.google.com/view/antonioplaza.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

From 1999 to 2023, he has been with Grenoble INP, where he was a Professor of signal and image processing. He is currently a Research Director with INRIA, Grenoble. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA; KTH, Stockholm, Sweden; and NUS, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavík, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot is a fellow of an ELLIS and an AAIA, a member of the Institut Universitaire de France from 2012 to 2017, and has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters), since 2018. He was the Founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRSS Chapter Excellence Award. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia from 2017 to 2019. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, and PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he served as a Guest Editor for the *IEEE Signal Processing Magazine*.



Danfeng Hong (Senior Member, IEEE) received the Dr.-Ing degree (summa cum laude) from the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, in 2019.

Since 2022, he has been a Full Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include artificial intelligence, multimodal vision, foundation models, and Earth science (AI4ES).

Dr. Hong has received several prestigious awards, including the Jose Bioucas Dias Award in 2021 and the Paul Gader Award in 2024 at WHISPERS for outstanding papers, respectively, the Remote Sensing Young Investigator Award in 2022, the IEEE GRSS Early Career Award in 2022, and the "2023 China's Intelligent Computing Innovators" Award (the only recipient in AI for Earth Science) by MIT Technology Review in 2024. He has been recognized as a Highly Cited Researcher by Clarivate Analytics in 2022–2024. He serves as an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP) and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS). He is also an Editorial Board Member of *Information Fusion*, the *ISPRS Journal of Photogrammetry and Remote Sensing*, and the *International Journal of Applied Earth Observation and Geoinformation*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:46:23 UTC from IEEE Xplore. Restrictions apply.